RATING METHODOLOGY

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Rating Methodology

Diversified Technology

This rating methodology replaces the *Diversified Technology* methodology published on 25 February 2022. While this methodology reflects many of the same core principles as the 2022 methodology, we have made changes to the scorecard to provide greater consistency of approach across our corporate sector methodologies. In the Scale factor, we have removed the EBIT subfactor. In the Profitability and Efficiency factor, we have replaced EBITDA Margin with EBIT Margin, and we have removed the Operating Income Return on Assets subfactor. In the Leverage and Coverage factor, we have replaced EBIT/Interest Expense with EBITDA/Interest Expense. We have changed some factor weights and scoring thresholds. We have also made editorial changes throughout the methodology to enhance readability, including expanding the "Other considerations" section for enhanced consistency across our corporate sector methodologies. We have also changed the presentation of the scorecard.

Scope

This rating methodology applies to companies globally that are primarily engaged in the design, manufacturing, and distribution of technology hardware and communication equipment products. The very broad diversified technology sector is highly fragmented and includes many subsectors such as personal computers, servers, storage and networking systems, smartphones and tablets and mailing and office equipment. Many companies that we rate using this methodology have diversified technology profiles, such that in addition to hardware, they generate revenue from software, services and captive finance operations.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

We rate technology companies that focus primarily on semiconductors, distribution, manufacturing, software or services using other rating methodologies.[1]

Sector overview

The highly competitive diversified technology sector includes companies that produce a wide range of products ranging from computers, servers, storage devices, communication equipment, hard disk drives, software, printers and copiers, smartphones, consumer electronics, and services. Some companies also undertake consulting activities and finance operations. The sector includes large diversified competitors, as well as smaller, specialized

companies. Many companies have diversified technology profiles, such that in addition to hardware, they generate revenue from software, services and captive finance operations.

Overall, the industry is characterized by uncertainties related to technology development; the pace of customer adoption of new products or services; periodic, intense pricing pressure; as well as ebbs and flows of the general economy. Companies with higher credit ratings tend to have diverse revenue streams or a high level of recurring revenue derived from services or supplies that follow an initial hardware sale.

Competition is also a defining feature for the industry. Throughout the various subsectors, competition generally evolves around technology, performance, price, quality, reliability, brand, distribution, customer service and support. Since product life cycles can be short, companies must continually invest in product enhancements as well as new products and services to remain competitive. Firms with strong balance sheets are able to continue investing during the down cycles. These firms are better positioned to address the sector's competitive demands and to gain market share through industry cycles.

Some of the rated debt relates to the captive finance operations of these companies. Finance operations enable diversified technology companies to provide financing options to customers and have been associated with companies in this sector that have had solid returns and consistent profitability. However, these operations also pose challenges for managing asset quality and liquidity. Finance operations are confidence sensitive because of their heavy reliance upon short-term financing. These operations reflect the usefulness of finance businesses in establishing and maintaining stronger customer relationships, particularly in the broader technology sector where financing can be an important part of an overall information technology solution for a customer.

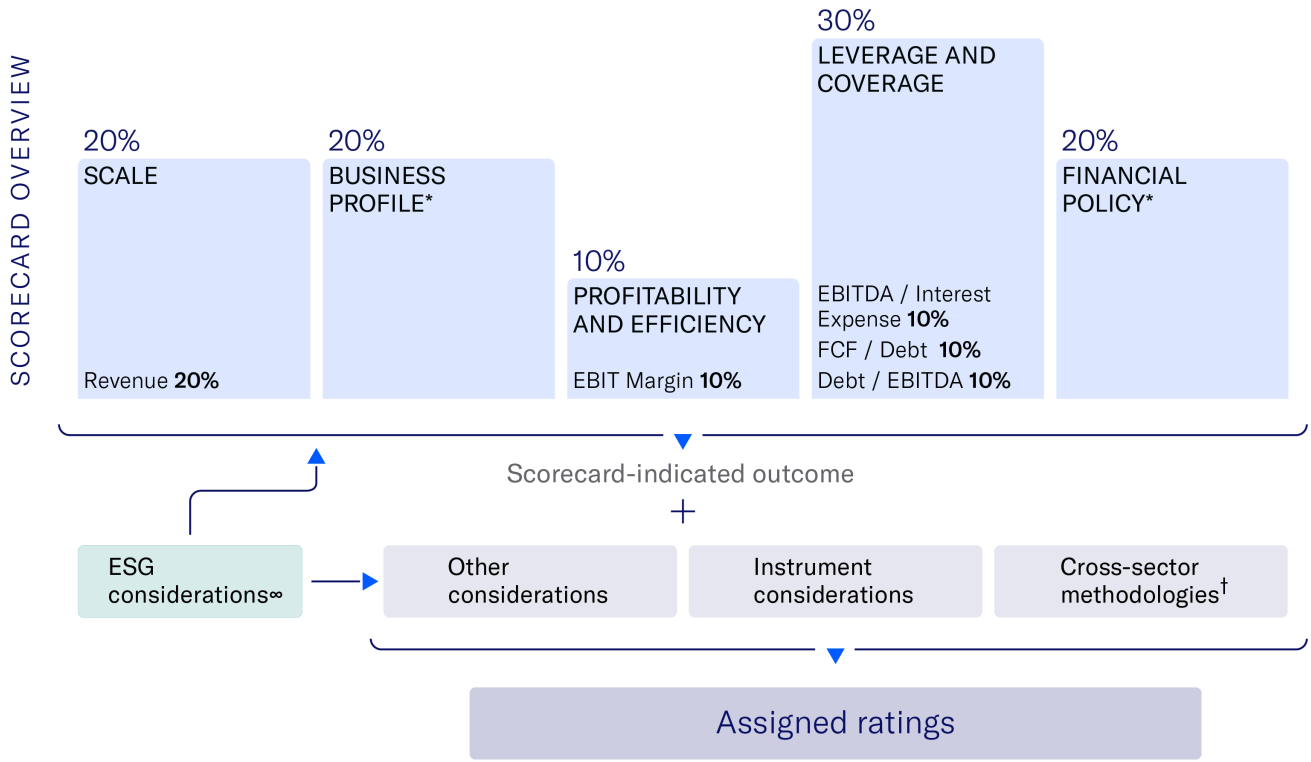
The captive finance operations of diversified finance companies typically do not file separate financial statements with the level of detailed exposure typical of companies that we rate using our captive finance company rating methodology.[2] However, their consolidation with the industrial parent complicates analytical comparison with the diversified technology companies that do not have similar subsidiaries. We make quantitative adjustments to financial metrics for companies with captive finance operations as a means to create a common basis for peer comparison, while also considering the incremental credit impact of these captive finance operations on the companies' overall creditworthiness (see appendix for details).

Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1
Illustration of the diversified technology rating approach



* This factor has no subfactors.
∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.
† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.
Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Diversified technology scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (20%)									
Revenue (USD Billion) [1]	20%	≥ $60	$30 - $60	$15 - $30	$8 - $15	$3 - $8	$1 - $3	$0.25 - $1	< $0.25
Factor: Business Profile (20%)									
Business Profile	20%	Expected volatility in results is almost non-existent. Supported by a commanding market position, entrenched cost effectiveness, technology advantages, and a well balanced global reach.	Very low expected volatility in results. Supported by a deeply entrenched and leading market position that is highly defensible through cost effectiveness and technology leadership with global exposure.	Low expected volatility in results. Supported by a strong market position, sustainable competitive advantages and solid diversity characteristics.	Moderate expected volatility in results. Supported by a solid market position with modest product differentiation. Good diversity characteristics provide a buffer against sudden shifts in demand.	Products are largely undifferentiated and the marketplace highly price competitive, exposing the company to periods of heightened volatility. Such exposure is tempered by an established market position and a favorable cost structure and fair diversity characteristics.	Products are undifferentiated, competition is intense and customers price sensitive, making results highly volatile. Company does not have advantageous cost profile or other competitive advantage to mitigate.	Results are expected to be extremely volatile. Company has modest market presence, few competitive advantages or may have above average costs.	Near term results are difficult to predict with any degree of confidence. No material competitive advantage.
Factor: Profitability and Efficiency (10%)									
EBIT Margin [2]	10%	≥ 40%	30% - 40%	20% - 30%	15% - 20%	10% - 15%	5% - 10%	2.5% - 5%	< 2.5%
Factor: Leverage and Coverage (30%)									
EBITDA / Interest Expense [3]	10%	≥ 30x	20x - 30x	10x - 20x	7x - 10x	4x - 7x	2x - 4x	1x - 2x	< 1x
FCF / Debt [4]	10%	≥ 45%	35% - 45%	25% - 35%	20% - 25%	10% - 20%	5% - 10%	0% - 5%	< 0%
Debt / EBITDA [5]	10%	≤ 0.5x	0.5x - 1x	1x - 1.5x	1.5x - 2.5x	2.5x - 4x	4x - 6x	6x - 8.5x	> 8.5x

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Financial Policy (20%)									
Financial Policy	20%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interest of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $100 billion. A value of $100 billion or better equates to a numeric score of 0.5. The Ca endpoint value is $0. A value of $0 equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is 70%. A value of 70% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5.
[3] For the linear scoring scale, the Aaa endpoint value is 40x. A value of 40x or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero or worse equates to a numeric score of 20.5.
[4] For the linear scoring scale, the Aaa endpoint value is 60%. A value of 60% or better equates to a numeric score of 0.5. The Ca endpoint value is (10)%. A value of (10)% or worse equates to a numeric score of 20.5.
[5] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 15x. A value of 15x or worse equates to a numeric score of 20.5, as does a negative EBITDA value.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[3] We may also make nonstandard adjustments that are specific to a particular company. (See the appendix for examples of nonstandard adjustments that apply to this sector.)

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Scale (20% weight)

Why it matters

Scale is an important indicator of the overall strength of a company's business, the size of its market and a company's relevance within an industry and the economy. Scale is also an indicator of a company's importance to its customers and bargaining strength with suppliers, as well as its ability to invest resources to maintain its competitive position and expand its marketing and geographic reach.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: Revenue.

Revenue

We use total reported revenue in billions of US dollars to measure scale, or to estimate it in the case of forward-looking expectations.

Factor: Business Profile (20% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

How we assess it for the scorecard

We score this factor based on our qualitative assessment of the durability of demand and a company's ability to meet that demand.

We assess the level of expected volatility in results and consider a company's market position and diversification. A company with both diversification and stability of market share typically receives a higher score for this factor. A strong position in one of these areas with weakness in the other is likely a limit to the long-term stability of cash flows.

Factor: Profitability and Efficiency (10% weight)

Why it matters

A company's profitability matters because it supports the generation of sustainable cash flow and reflects the strength of the company's competitive position. Profitability also provides insights into a company's ability to manage key costs and pass them on to customers, which is an important indicator of the long-term viability of its business model. Strong profitability and operating efficiency can help a company to withstand economic and cyclical downturns that affect its competitive position and credit metrics.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: EBIT Margin.[4]

EBIT Margin

We use the ratio of earnings before interest and taxes (EBIT) to revenue.

Factor: Leverage and Coverage (30% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on three quantitative subfactors: EBITDA/Interest Expense; FCF/Debt and Debt/EBITDA.[5]

EBITDA / Interest Expense

We use the ratio of earnings before interest, taxes, depreciation and amortization (EBITDA) to interest expense.

FCF / Debt

We use the ratio of free cash flow (FCF) to total debt.

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

Factor: Financial Policy (20% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[6]

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[7]

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we typically consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics in the scorecard or in other rating considerations.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[8]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[9]

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 3

Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 4

Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[10]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[11]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[12]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[13] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Appendix: Nonstandard adjustments to financial statements

In addition to making standard adjustments to a company's financial statements, we may also make nonstandard adjustments to better reflect underlying economics, improve comparability with peer companies and reflect estimates or assumptions we consider to be more appropriate for credit analysis.

We make nonstandard financial statement adjustments based on public information, although we may consider private information in our ratings. We may apply a materiality threshold to determine whether or not to make a nonstandard adjustment to a particular company's financial statements.

In this sector, captive finance operations are fundamentally different from nonfinance operations in many respects, including margin structure, interest coverage, capital adequacy, returns on assets and cash flow generation. For a company with captive finance operations, we typically evaluate the finance business by considering profitability, returns, leverage, liquidity, asset quality and operational scale. In addition to these quantitative elements, we assess the strategic importance of the company's finance business to the parent's overall strategy. Our overall assessment of the impact of a finance business on a diversified technology company's credit profile is informed by quantitative and qualitative elements.

We typically make adjustments for diversified technology companies with captive finance operations as follows:

» We adjust the balance sheet to eliminate our estimate of finance-related assets, debt and equity. We generally assume a seven times debt-to-equity ratio for net finance assets. Information for this adjustment is typically based on public disclosures regarding finance assets, which permit us to estimate the mix of debt and equity related to the finance business.

» We adjust the income statement to eliminate income that we estimate is attributable to finance operations.

» Using reported or estimated changes in finance receivables and in equipment on operating leases, as well as for estimated net income of captive finance operations, we adjust cash flow from operations by moving these changes to the investing section of the statement of cash flows.

The specific adjustments discussed below are generally applied unless we have financial or other information regarding a particular company's finance operations that would cause us to believe that a different approach is more appropriate. Although there is variation in the composition of finance assets, liabilities and profitability across the sector, the typical adjustments discussed below provide a framework for comparing results across the sector.

Adjustments to debt and leverage

The capital structure of companies with captive finance operations, consistent with pure play finance companies, typically employs a higher mix of debt financing (relative to that used by nonfinance operations) to support and fund finance assets. To adjust for finance operations as a means to normalize debt and equity levels, we typically assume a seven times debt-to-equity level for finance assets.

In contrast to a manufacturer, debt supporting finance assets is repaid by a reduction of the finance assets/receivables. Hence, to better evaluate and compare the nonfinance business of these diversified technology companies to peer companies without financing operations, we remove the finance receivables from the balance sheet assets, as well as debt and equity that are allocated to the finance business.

Adjustments to the income statement

We adjust the income statement for income that we estimate is attributable to finance operations. We typically assume that the captive finance companies earn a 4% pretax return on average assets and multiply the average of beginning-of-period and end-of-period finance assets by 4% to derive an estimate for finance company pretax income. We reduce each company's quarterly and annual earnings by the respective amount of quarterly and annual finance pretax income by adjusting cost of goods sold.

Adjustments to cash flow

We adjust the statement of cash flows for changes in finance receivables and equipment on operating leases as well as for estimated net income of captive finance operations. We generally do not make any other adjustments for items such as deferred taxes attributed to finance operations.

Our approach considers that finance assets are an investment to generate profit, and not working capital that should be minimized for efficiency (as with a traditional manufacturer). We therefore move changes in finance receivables and equipment on operating leases from the operating section in the cash flow statement to the investing section. With this adjustment, increases in finance receivables (often a reflection of higher sales, which is generally favorable for credit quality, subject to asset quality) would not negatively affect core cash flow from operations. Decreases in finance receivables (often not favorable for credit quality in this sector as it may reflect weaker product competitiveness) would not positively affect core cash flow from operations. We also move the net income that we estimate is attributable to finance operations into the investing section of the statement of cash flows because the net income helps to support growth in finance operations.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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6 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

12 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

13 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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